MODEL DESIGN DOCUMENT

Sales and Profit Analysis Using SQL and Tableau

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1. Executive summary

This project presents a complete sales and profit analysis workflow built using SQL and Tableau. The work begins with raw sales transaction data, which is first protected through backup copies, then explored, cleaned, structured, and transformed into analysis-ready tables. The final analytical layer is used to generate business metrics such as revenue, sales quantity, profit, profit margin, market-wise performance, customer contribution, and monthly trends.

The project is organized into two related analytical parts: a revenue analysis flow and a profit analysis flow. The revenue analysis is kept in the original sales structure, while the profit analysis is built in a separate schema so that both can be presented independently without affecting each other. The final output is presented through interactive dashboards that convert raw transactional data into meaningful business insights.

2. Dataset Overview

2.1 Dataset Origin

The dataset used in this project is a sample sales dataset commonly used for data analytics practice and business intelligence projects. The dataset represents a fictional company selling products across multiple markets and customers over a period of time. It contains transaction data along with customer, product, market, and date information, which makes it suitable for SQL analysis, data modelling, and dashboard development.

2.2 Dataset Summary

This project uses a structured sales database containing **five related tables**. The `transactions` table acts as the main fact table, while the `customers`, `products`, `markets`, and `date` tables act as supporting dimension tables. This structure allows the data to be analyzed across time, market, customer, and product dimensions.

2.3 Key Features of the Dataset

Feature	Description
Total number of tables	5
Main fact table	`transactions`
Dimension tables	`customers`, `products`, `markets`, `date`
Total transaction rows	1,50,281
Customer rows	38
Product rows	279

Market rows	17
Date rows	1,126
Time period covered	2017 to 2020
Customer types	Brick & Mortar, E-Commerce
Key fields	`customer_code`, `product_code`, `market_code`, `order_date`
Business measures	`sales_qty`, `sales_amount`, `currency`
Descriptive fields	`custmer_name`, `customer_type`, `product_type`, `markets_name`, `zone`, `year`, `month_name`, `date_yy_mmm`

The dataset is well suited for analysis because it contains both transaction measures and descriptive lookup tables, which makes it ideal for star schema modelling and dashboard creation.

2.4 Why the Dataset Matters

This dataset is useful for business analysis because it allows performance to be studied from multiple angles, such as revenue, sales quantity, profit, customer contribution, market performance, and time trends. Since the data is already organized into transaction and dimension tables, it is easy to transform it into a reporting-friendly model for SQL analysis and Tableau dashboards.

3. Project Overview

3.1 Background

In modern businesses, sales data is one of the most important sources of decision-making. Every transaction contains useful information about customers, markets, products, and time-based performance. However, raw data by itself is not easy to interpret. It must be cleaned, structured, and modelled properly before it can support analysis and dashboarding.

This project focuses on turning raw sales data into a business intelligence solution. The goal is not only to show sales numbers but also to understand profitability, customer behaviour, market performance, and sales trends over time. By using SQL for data preparation and Tableau for visualisation, the project demonstrates a practical end-to-end analytics workflow.

3.2 Business Use Case

A business may have large volumes of sales records, but it still may not know:

- which markets are most profitable,
- which customers contribute the highest revenue,
- which products perform consistently,
- whether sales growth is matched by profit growth,
- and whether the business is too dependent on a few markets or customers.

This project solves that problem by organising the data into a clean analytical structure and building separate dashboards for revenue and profit analysis. The result is a model that can support better business decisions.

3.3 Problem Statement

Raw transactional data often contains inconsistencies, duplicate values, missing values, and mixed formats. In addition, revenue alone does not give a full picture of business performance because high revenue does not always mean high profit.

The challenge is to transform raw sales transactions into a reliable analytical structure that can explain both revenue and profitability patterns across customers, products, markets, and time.

3.4 Objectives

The main objectives of this project are:

- to understand the structure of the sales dataset,
- to preserve raw data safely before any changes are made,
- to clean and prepare the data for analysis,
- to build a star schema for efficient reporting,
- to create final analytical tables for dashboard use,
- to calculate and analyse revenue, sales quantity, profit, and profit margin,
- to identify market-wise, customer-wise, and product-wise performance,
- to build interactive dashboards in Tableau,
- and to generate useful business insights and recommendations.

3.5 Scope

The scope of this project includes:

- data overview and profiling,
- raw data backup,
- data cleaning and transformation,

- star schema creation,
- SQL-based revenue and profit analysis,
- interactive dashboard creation,
- and insight generation.

The project focuses on descriptive analytics and dashboarding. It does not cover forecasting or machine learning as part of the final dashboard workflow.

4. Tools Used

This project was completed using the following tools:

- **MySQL / MySQL Workbench**: Used for importing the dataset, checking table structure, creating backup copies, cleaning the data, building the star schema, and writing SQL analysis queries.
- **SQL**: Used for data exploration, data cleaning, joins, aggregations, and final business analysis.
- **Tableau Public**: Used for creating interactive dashboards and visualising the results.
- **CSV Export**: Used to move the final analytical table from MySQL into Tableau Public.

5. Data Understanding and Data Overview

Before cleaning the data, the structure of the dataset was studied carefully.

5.1 Data Overview Checks

The overview stage includes:

- table structure checks,
- row count checks,
- date range checks,
- null value checks,
- and category distribution checks.

This stage helps identify whether the data is complete, consistent, and ready for transformation. The SQL scripts used in your project already reflect this approach through table descriptions, row counts, date range queries, distinct-value checks, null checks, and duplicate verification.

5.2 Observations from the Data

During exploration, the dataset was understood as a business database with:

- transaction records,
- customer master information,
- product master information,
- market information,
- and date information.

This structure is suitable for a dimensional model because the transaction table can act as the fact table while the other tables provide descriptive context.

6. Data Preparation and Preprocessing

6.1 Why Preprocessing Was Needed

Raw data cannot be directly used for dashboarding because it may contain invalid values, inconsistent formatting, or repeated records. Preprocessing ensures that the final analytical output is accurate, trustworthy, and easy to use.

6.2 Backup Strategy

Before making any edits, the original tables were kept safe and backup copies were created. This protects the source data and makes it possible to recover the original state if needed.

In your workflow, the copy tables were created first and then used for cleaning and analysis. This is a good practice because it separates raw storage from transformation work.

6.3 Cleaning Tasks Performed

The preprocessing stage included:

- checking for zero and negative sales values,
- checking null values,
- checking currency consistency,
- checking market and product distributions,
- checking duplicate rows in lookup tables,
- standardising text fields,
- and preparing the data for joins.

This is the same kind of structure seen in strong academic MDDs: a clear transformation phase before EDA and visualisation.

6.4 Analytical Preparation

After cleaning, the data was prepared in a way that supports reporting:

- a final analytical table was created for revenue analysis,
- a separate profit table was prepared for profitability analysis,
- and the data was organized for Tableau export.

This final layer reduces complexity at the visualisation stage and makes the workbook easier to manage.

7. Data Modeling

7.1 Why a Star Schema Was Used

A star schema makes analysis easier because it separates numeric measures from descriptive attributes. The fact table stores transaction or performance values, while the dimension tables store lookup information such as customer names, market names, product types, and dates.

This structure is ideal for dashboards because it supports simple and fast aggregation. It is also easier to explain in a presentation because the model is clean and business-friendly.

7.2 Tables in the Model

The model is based on:

- one fact table,
- and multiple dimension tables.

Fact table

Contains:

- sales quantity,
- sales amount,
- profit,
- and related transaction measures.

Dimension tables

Contain:

- customer details,
- product details,

- market details,
- and date details.

The SQL files in your project already show this modeling step through separate scripts for customer, product, market, date, fact table, final analysis table, SQL analysis, and export verification.

7.3 Benefit of the Model

The star schema allows the final dashboard to answer questions such as:

- which market contributes the most revenue,
- which market has the best profit margin,
- which customers are most valuable,
- and how performance changes over time.

8. SQL Analysis and Business Measures

8.1 Core Metrics

The following measures were calculated through SQL:

- total revenue,
- total sales quantity,
- total profit,
- profit margin percentage,
- revenue by market,
- profit margin by market,
- revenue by year,
- monthly trend,
- top customers,
- top products,
- and customer type contribution.

8.2 Why SQL Was Used

SQL was used because it is efficient for:

- filtering,
- joining,
- grouping,
- aggregating,
- and preparing final data for export.

Instead of performing all logic inside Tableau, the most important calculations were handled in the database layer first. This makes the dashboard cleaner and more stable.

8.3 Final Analytical Table

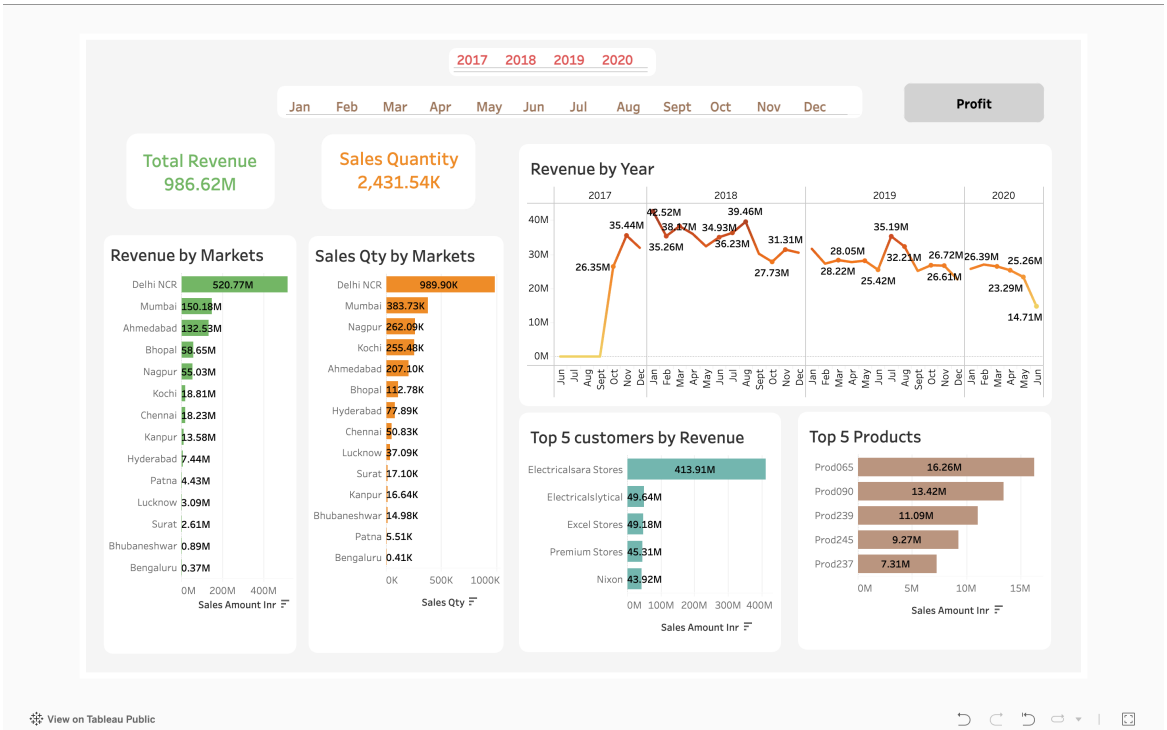
The final analytical table includes all important dimensions and measures in one place. This table is the main source for dashboarding because it already contains:

- customer details,
- product details,
- market details,
- date details,
- sales measures,
- and profit measures.

That makes the Tableau workbook easier to design and less dependent on complex joins.

9. Dashboard Design

9.1 Revenue Dashboard



The revenue dashboard was created to visualise sales performance.

It includes:

- Total Revenue KPI,
- Sales Quantity KPI,
- Revenue by Markets,
- Sales Quantity by Markets,

- Revenue by Year,
- Top 5 Customers,
- Top 5 Products.

This dashboard shows where revenue is coming from and which segments contribute most to the business.

9.2 Profit Dashboard

The profit dashboard was created to show profitability, not just sales.

It includes:

- Total Revenue KPI,
- Total Profit KPI,
- Profit Margin KPI,
- Revenue by Markets,
- Profit Margin by Markets,
- Profit Trend,
- Customer Table,
- Customer Type Pie Chart.

This dashboard helps answer a more important business question: not just how much was sold, but how efficiently the business earned profit.

9.3 Filters and Navigation

The dashboards use:

- year filters,
- month filters,
- and navigation buttons between the revenue and profit pages.

This allows the user to move between views easily without losing the context of the analysis.

10. Key Insights

The dashboards reveal several important findings based on the revenue and profit analysis:

- **Delhi NCR is the highest revenue market**, contributing **520.77M** in revenue, far ahead of all other markets. Mumbai and Ahmedabad follow at **150.18M** and **132.53M** respectively.
- **Sales quantity is also dominated by Delhi NCR**, which records **989.90K** units sold, showing that it is the most active market in terms of transaction volume as well.
- **Revenue and profit margin do not follow the same pattern.** Some markets with strong revenue still have lower profit margins, which means high sales volume does not always lead to high profitability. On the profit dashboard, **Surat** has the highest profit margin at **4.86%**, while **Bengaluru** shows a negative margin of **-20.78%**.
- **Most of the business comes from Brick & Mortar customers**, which account for **75.60%** of the customer type split, while **E-Commerce** contributes **24.40%**.
- **A small group of customers contributes a large share of revenue.** The top customer, **Electricalsara Stores**, contributes **413.91M**, which is much higher than the other top customers shown in the dashboard.
- **Product revenue is concentrated in a few products.** The top product, **Prod065**, generates **16.26M**, followed by **Prod090** at **13.42M** and **Prod239** at **11.09M**.
- **Profitability is more uneven than revenue.** While overall revenue is high, the total profit is **24.66M**, which shows that margin management is important and should be analyzed along with sales volume.
- **The trend charts show that performance changes over time**, so sales and profit should be monitored monthly and yearly rather than only at the total level.

These insights show that the business should not focus only on revenue, but must also compare revenue with profit margin to understand true performance.

11. Business Recommendations

Based on the analysis of revenue, sales quantity, and profit margin across markets, customers, products, and time, the following business recommendations can be made:

1. **Focus on High-Revenue Markets**
   Delhi NCR, Mumbai, and Ahmedabad generate the highest revenue. The company should focus on maintaining strong relationships, marketing efforts, and supply chain efficiency in these markets because they are the main revenue drivers.
2. **Improve Profit Margin in Low-Margin Markets**
   Some markets generate revenue but have low or negative profit margins, such as Bengaluru. The company should review pricing strategy, transportation cost, and operational expenses in these markets to improve profitability.
3. **Increase Focus on High-Margin Markets**
   Markets like Surat, Patna, and Bhubaneswar show higher profit margins. The company should consider expanding operations and sales efforts in these markets because they generate profit more efficiently.
4. **Customer Concentration Risk Management**
   A large portion of revenue comes from a few top customers such as Electricalsara Stores. The business should maintain strong relationships with these customers but also try to diversify the customer base to reduce dependency on a small number of customers.
5. **Product Performance Monitoring**
   A few products contribute most of the revenue. The company should ensure continuous availability of top-performing products and analyze why some products perform better than others.
6. **Monitor Profit Along With Revenue**
   High revenue does not always mean high profit. The company should track profit margin along with revenue before making business decisions.
7. **Customer Type Strategy**
   Since most revenue comes from Brick & Mortar customers, the company should maintain strong offline sales channels while also trying to grow the E-Commerce segment for future growth.
8. **Use Time-Based Analysis for Planning**
   Since sales and profit change over months and years, the company should use monthly and yearly trend analysis for forecasting, inventory planning, and sales target setting.

12. Limitations and Future Scope

Limitations

This project has the following limitations:

- The analysis is based only on historical data from 2017 to 2020.
- The project focuses on descriptive analytics (what happened), not predictive analytics (what will happen).

- Profit is calculated using the available cost and sales data; if cost data changes, profit values may also change.
- The dashboards are static and require manual data refresh because Tableau Public does not support live database connections.
- The analysis is limited to available fields such as sales amount, quantity, market, customer, and product. Other business factors like marketing spend, discounts, and logistics cost are not included in the dataset.

Future Scope

This project can be extended in the future in the following ways:

- Sales and profit forecasting can be added using time series models.
- Customer segmentation can be performed using RFM (Recency, Frequency, Monetary) analysis.
- Product category level profitability analysis can be added.
- A live dashboard can be built using Tableau Server or Power BI with automatic data refresh.
- A complete ETL pipeline can be created to automate data cleaning and transformation.
- Additional KPIs such as average order value, customer lifetime value, and growth rate can be added.

13. Conclusion

This project demonstrates how raw transactional data can be transformed into useful business information through structured SQL processing and Tableau dashboarding. The data was first understood and protected, then cleaned and modeled into a reporting-friendly structure. After that, revenue and profit analysis were carried out, and the results were displayed in interactive dashboards.

The final outcome is a complete business analytics workflow that shows not only how much the company sold, but also how profitable those sales were across customers, markets, products, and time periods.